

Planning algorithms with PDDL language

Artificial Intelligence and Knowledge Engineering – Lis No. 3

Goal

The goal of this list is to get acquainted with the PDDL language (Planning Domain Definition Language) developed to standardize Artificial Intelligence (AI) planning languages. This list consists of three tasks with different planning problems to solve.

The lab report should include a description of a problem, a screenshot a generated plan, analysis of experiments, conclusions and .pddl files.

How to run PDDL?

You can use a planner like Fast Downward or an online editor:

- `fast-downward.py domain.pddl problem.pddl - search "astar(lmcut)"`
- Edytor online: <https://editor.planning.domains>

Task 1 (25 pts)

Design a package transport plant using the PDDL language with extensions. Develop a logical model and implement it as PDDL files, then run, test it and evaluate the generated plan. Analyze the duration of the plan, its cost, and influence of transport topology. The following PDDL extensions may be used in the project:

- `:strips` – basic STRIPS operation model,
- `:typing` – object typing (e.g. `package`, `location`, `vehicle`),
- `:negative-preconditions` – negative conditions (e.g. `(not (at package location))`),
- `:conditional-effects` – optional conditional effects,
- `:multi-agent` – for an extended model with multiple vehicles,
- action costs (`:numeric-fluents`, `:action-costs`),
- duration (`:durative-actions`),
- complex transport topology (road, air and water).

Task 2 (15 pts)

Develop a plan in PDDL language in which a robot visits all rooms and cleans them. Write the `domain.pddl` and `problem.pddl` files. Run a planner and analyze the generated plan.

- Objects: robot: `robot`, rooms: `room1`, `room2`, `room3`.
- Predicates:
 - `(at ?r - robot ?p - room)` – robot is in the room,
 - `(dirty ?p - room)` – the room is dirty,

- (clean ?p - room) – room is clean.
- Actions:
 - move – moves the robot between rooms;
 - clean – cleans the current room.
- Goal: All rooms are clean: (and (clean pokoj1) (clean pokoj2) (clean pokoj3))

Zadanie 3 (10 pkt)

Basing on the lecture and informations from this page www.ida.liu.se, solve the following task. Run the provided PDDL model, generate a plan and analyze it.

Task description: The robot has two arms and can move between two rooms. The first room contains four balls and the robot itself. The goal is to move all the balls to the second room using the actions of picking up and putting down the balls.

Objects:

- Objects: room1, room2,
- Balls: ball1, ball2, ball3, ball4,
- Robot's arms: arm1, arm2,
- Robot: robot.

Predicates:

- (at ?r - robot ?room - room) – the robot is in the room;
- (inroom ?b - ball ?room - room) the ball is in the room;
- (holding ?a - arm ?b - ball) – arm holds the ball;
- (arm-empty ?a - arm) – arm is empty.

File domain.pddl

```

1 (define (domain ball-moving-robot )
2   (: requirements : strips : typing )
3   (: types robot room ball arm)
4
5   (: predicates
6     (at ?r - robot ?rm - room )
7     ( inroom ?b - ball ?rm - room )
8     ( holding ?a - arm ?b - ball )
9     ( arm-empty ?a - arm )
10  )
11
12  (: action move
13    : parameters (?r - robot ? from - room ?to - room )
14    : precondition (at ?r ? from )
15    : effect (and (not (at ?r ? from )) (at ?r ?to))
16  )
17
18  (: action pick-up
19    : parameters (?r - robot ?a - arm ?b - ball ?rm - room )
20    : precondition ( and (at ?r ?rm) ( inroom ?b ?rm) ( arm-empty ?a))
21    : effect (and ( holding ?a ?b) (not ( arm-empty ?a)) (not ( inroom ?b ?rm)))
22  )
23
24  (: action put-down
25    : parameters (?r - robot ?a - arm ?b - ball ?rm - room )
26    : precondition ( and (at ?r ?rm) ( holding ?a ?b))
27    : effect (and ( inroom ?b ?rm) ( arm-empty ?a) ( not ( holding ?a ?b)))
28  )
29 )

```

File problem.pddl

```
1 define ( problem move-balls )
2   (: domain ball-moving-robot )
3
4   (: objects
5     room1 room2 - room
6     robot - robot
7     ball1 ball2 ball3 ball4 - ball
8     arm1 arm2 - arm
9   )
10
11   (: init
12     (at robot room1 )
13     ( inroom ball1 room1 )
14     ( inroom ball2 room1 )
15     ( inroom ball3 room1 )
16     ( inroom ball4 room1 )
17     ( arm-empty arm1 )
18     ( arm-empty arm2 )
19   )
20
21   (: goal
22     (and
23       ( inroom ball1 room2 )
24       ( inroom ball2 room2 )
25       ( inroom ball3 room2 )
26       ( inroom ball4 room2 )
27     )
28   )
29 )
```

Additional materials

- <http://editor.planning.domains>
- <https://www.fast-downward.org/>
- <https://planning.wiki/ref/pddl/domain>
- <https://fareskalaboud.github.io/LearnPDDL/>
- <https://www.cs.toronto.edu/~sheila/2542/s14/A1/introtopddl2.pdf>
- http://pddl4j.imag.fr/pddl_tutorial.html